Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following statements regarding Hg^{2+} (ionic radius $116pm$) and Ca^{2+} (ionic radius $114pm$) is correct?

(A) Hg^{2+} ion has more polarizing effect on an anion as compared to Ca^{2+} ion

(B) Hg^{2+} ion has less polarizing effect on an anion as compared to Ca^{2+} ion

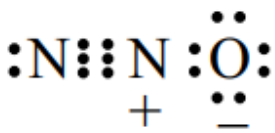
(C) Both Hg^{2+} ion and Ca^{2+} have equal polarizing effect on an anion

(D) Both Hg^{2+} ion and Ca^{2+} have no polarizing effect on an anion

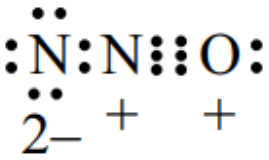
Q.2

Which of the following resonating structures of N_2O is feasible in its linear structure?

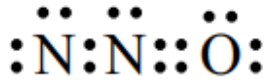
(A)



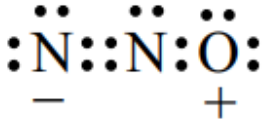
(B)



(C)



(D)



Q.3

Which of the following pairs of species are not isostructural?

(A) XeF_4, PF_4^+

(B) CH_4 and NH_4^+

(C) XeF_2, I_3^-

(D) XeF_5^+, IF_5

Q.4

Which of the following facts regarding Fajan's rules is not correct?

(A) A small positive ion favours covalency

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(B) A large negative ions favours covalency

(C) Large charge on either ion, or on both ions, favours covalency

(D) Covalency is favoured if the polarizing ion has a noble gas configuration

Q.5

Which of the following pairs of species has identical value of bond order?

(A) NO^+, N_2

(B) CO and N_2^+

(C) CN^-, O_2^+

(D) N_2 and O_2^+

Q.6

Which one of the following is expected to have maximum bond length?

(A) NO^+

(B) NO^{2+}

(C) NO

(D) NO^-

Q.7

Which one of the following species will have maximum bond dissociation energy?

(A) C_2

(B) C_2^+

(C) C_2^-

(D) C_2^{2-}

Q.8

Which of the following statements regarding carbon monoxide is incorrect?

(A) A triple bond exists between C and O

(B) O atom carries negative charge and C carries positive charge

(C) Dipole moment of CO is low as compared to its calculated value from the charge separate and bond distance

(D) Bond order of CO is 3.

Q.9

Which of the following interactions is not a part of van der Waals interactions?

(A) dipole-dipole interactions

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(B) dipole-induced dipole interactions

(C) dispersion forces

(D) ion-dipole interactions

Q.10

Which of the following facts regarding ClF_3 is correct?

(A) All the three $Cl - F$ bond has the same length

(B) The angles $F - Cl - F$ are 120°

(C) There is no lone pair of electrons around Cl

(D) There are two lone pair of electrons occupying equatorial positions

Q.11

In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed?

(A) $C_2 \rightarrow C_2^+$

(B) $NO \rightarrow NO^+$

(C) $O_2 \rightarrow O_2^+$

(D) $N_2 \rightarrow N_2^+$

Q.12

Based on lattice energy and other consideration which one of the following alkali metal chlorides is expected to have the highest melting point?

(A) KCl

(B) $RbCl$

(C) $LiCl$

(D) $NaCl$

Q.13

The hybridization of iodine orbitals in the compound IF_7 is

(A) $sp^3\ d$

(B) $sp^3\ d^2$

(C) $sp^2\ d^3$

(D) $sp^3\ d^3$

Q.14

Lattice energy of an ionic compound depends upon

(A) packing of ions only

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(B) charge and size of the ions

(C) charge on the ion only

(D) size of the ion only

Q.15

The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, $K^+, Ca^{2+}, Mg^{2+}, Be^{2+}$?

(A) $Mg^{2+} < Be^{2+} < K^+ < Ca^{2+}$

(B) $Be^{2+} < K^+ < Ca^{2+} < Mg^{2+}$

(C) $K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$

(D) $Ca^{2+} < Mg^{2+} < Be^{2+} < K^+$

Q.16

Which of the following statements regarding bond angles in the given molecules is correct?

(A) $NF_3 > NH_3$ and $OF_2 > OH_2$

(B) $NF_3 > NH_3$ and $OF_2 < OH_2$

(C) $NF_3 < NH_3$ and $OF_2 > OH_2$

(D) $NF_3 < NH_3$ and $OF_2 < OH_2$

Q.17

The geometry of $XeOF_4$ by VSEPR theory is:

(A) trigonal bipyramidal

(B) square pyramidal

(C) octahedral

(D) pentagonal planar

Q.18

Which of the following statement is not correct?

(A) Hybridization involves the mixing of atomic orbitals of the same atom.

(B) Hybridization of orbitals results into the formation of equivalent orbitals in all respect.

(C) Hybridization may involve any combination of s, p and d orbitals.

(D) Hybridization of orbitals predicts the molecular geometry of the molecule and not vice-versa.

Q.19

Which of the following orders is correct for the bond dissociation energy of O_2, O_2^+, O_2^- and O_2^{2-}?

(A) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

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(B) $O_2^+ > O_2 < O_2^- < O_2^{2-}$

(C) $O_2^+ < O_2 < O_2^- < O_2^{2-}$

(D) $O_2^+ > O_2 > O_2^- < O_2^{2-}$

Q.20

Which of the following statements is not correct from the viewpoint of molecular orbital?

(A) Be_2 is not a stable molecule

(B) He_2 is not stable but He_2^+ is expected to exist

(C) Bond strength of N_2 is maximum amongst the homonuclear diatomic molecules belonging to the second period

(D) The order of energies of molecular orbitals in F_2 molecule is

$$E(\sigma 2\,s) < E\,(\sigma^* 2\,s) < E\,(\pi 2p_x) = E\,(\pi 2p_y) < E(\sigma 2p) < E\,(\pi^* 2p_x) = E\,(\pi^* 2p_y) < E\,(\sigma^* 2p)$$

Q.21

The number of lone pairs of electrons around I in IF_3, IF_5 and IF_7, respectively. Find $IF_3 + IF_5$ - IF_7.

Q.22

The number of bonding and antibonding electrons in O_2^{2-} ion, respectively. Find bonding electrons $-$ antibonding electrons.

Q.23

The bond order of CN^- is

Q.24

The dipole moment of hydrogen chloride bond distance: $127pm$ is $1.03D$. The per cent ionic character of its bond is about. (Mark answer to nearest integer)

Q.25

Molecule AB has a bond length of $1.617\mathring{A}$ and a dipole moment of $0.38D$. The fractional charge on each atom (absolute magnitude) is: $\left(e_0 = 4.802 \times 10^{-10}\right.$ esu). (Round off answer to two decimal places)

Q.26

Maximum number of atoms present in one plane in $PF_2\,(CH_3)_3$.

Q.27

How many of these statements are correct ?
(I) $I_3^{\ominus}$ has bent structure.

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(II) The number of $p\pi - p\pi$ and $d\pi - p\pi$ bonds present in SO_2 is 1.
(III) $XeF_5^{\ominus}$ has pentagonal planar shape but PCl_5 is trigonal bipyramidal.

Q.28

The maximum number of hydrogen bonds that can be formed by water molecules is

Q.29

Total number of $p\pi - d\pi$ bonds in XeO_2F_2

Q.30

The bond distance of $S - O$ in SO_2 is $145pm$ and its bond moment is $5.8 \times 10^{-30}Cm$. Its per cent ionic character is about (Mark answer to nearest integer)

Answers & Solutions

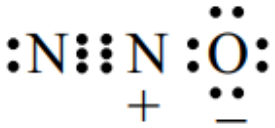
Q.1 Answer:

Hg^{2+} ion has more polarizing effect on an anion as compared to Ca^{2+} ion

Solution:

Hg^{2+} ion belongs to transition metal and Ca^{2+} ion to alkaline-earth metal. Thus Hg^{2+} ion has more polarizing effect on an anion.

Q.2 Answer:



Solution:

Structure a is feasible. In the choices (b) and (d) +ve charges is placed on the more electronegative oxygen. Also, the structures (c) and (d) are expected to be angular.

Q.3 Answer:

XeF_4, PF_4^+

Solution:

XeF_4 has 6 pairs of electrons around Xe while PF_4^+ has 4 pairs of electrons around P.

Q.4 Answer:

Covalency is favoured if the polarizing ion has a noble gas configuration

Solution:

A noble gas configuration is the most effective at shielding the nuclear charge and has a poor polarizing effect.

Q.5 Answer:

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NO^+, N_2

Solution:

Valence electrons of $N_2 = 10$ **Electronic Configuration** $(\sigma 2\,s)^2(\sigma * 2\,s)^2\,(\pi 2p_x)^2\,(\pi 2p_y)^2\,(\sigma 2p_z)^2$
Bond order $= (8-2)/2 = 3$
Valence electrons of $NO^+ = 10$ $(\sigma 2\,s)^2(\sigma * 2\,s)^2\,(\pi 2p_x)^2\,(\pi 2p_y)^2\,(\sigma 2p_z)^2$
Bond order $= (8-2)/2 = 3$
Valence electrons of $CO = 10$; **Same as** NO^+
Valence electrons of $CN^- = 10$; **Same as** NO^+
Valence electrons of $N_2^- = 11$ **Bond order** $= (9-2)/2 = 3.5$
Valence electrons of $O_2^+ = 11$ **Electronic configuration** $(\sigma 2\,s)^2(\sigma * 2\,s)^2\,(\sigma 2p_z)^2\,(\pi 2p_x)^2\,(\pi 2p_y)^2\,(\pi^* 2p_x)^1$
Bond order $= (8-3)/2 = 2.5$

Q.6 Answer:

NO^-

Solution:

Electronic configuration follows the order $(\sigma 2\,s)(\sigma * 2\,s)\,(\pi 2p_x)\,(\pi 2p_y)\,(\sigma 2p_z)\,(\pi^* 2p_x)$
NO^+ **(Valence electrons=10); Bond order** $= (8-2)/2 = 3$
NO^{2+} **(Valence electrons=9); Bond order** $= (7-2)/2 = 2.5$
NO **(Valence electrons=11); Bond order** $= (8-3)/2 = 2.5$
NO^- **(Valence electrons=12); Bond order** $= (8-4)/2 = 2.0$
Smaller the bond order, larger the bond length. Hence, NO^- **will having maximum bond length.**

Q.7 Answer:

C_2^{2-}

Solution:

C_2**(8 electrons);** $KK(\sigma 2\,s)^2(\sigma * 2\,s)^2\,(\pi * 2p_x)^2\,(\pi 2p_y)^2$ **Bond order** $= (6-2)/2 = 2$
C_2^+**(7 electrons);** **Bond order** $= (5-2)/2 = 1^1/2$
C_2^- **(9 electrons);** $KK(\sigma 2\,s)^2(\sigma * 2\,s)^2\,(\pi 2p_x)^2\,(\pi 2p_y)^2\,(\sigma 2p_y)^1$ **Bond order** $= (7-2)/2 = 2^1/2$
C_2^{2-}**(10 electrons);** **Bond order** $= (8-2)/2 = 3$
Larger the bond order, larger the dissociation energy.

Q.8 Answer:

O atom carries negative charge and C carries positive charge

Solution:

The CO molecule is represented as : $\overline{C} \leftrightarrows \overset{+}{O}$ **:**
Dipole moment is low due to pull of bonding electrons towards more electronegative oxygen atom.

Q.9 Answer:

ion-dipole interactions

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Solution:

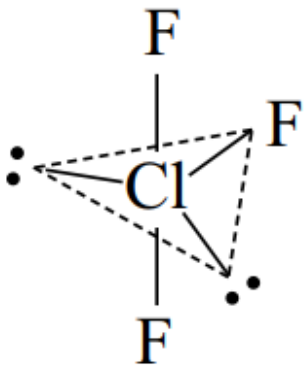
Ion-dipole interaction is not counted in van der Waals forces.

Q.10 Answer:

There are two lone pair of electrons occupying equatorial positions

Solution:

The structure of ClF_3 is



The F-Cl-F bond angles is 87°. It is less than 90° because of lone-pair-bond pair electron repulsions. The two bond lengths are $169.8\mathrm{pm}$ (axial bond) and $159.8\mathrm{pm}$ (equatorial bond)

Q.11 Answer:

$NO \rightarrow NO^{+}$

Solution:

The order of increasing energy of molecular orbitals is

$$(\sigma 2\,\mathrm{s})(\sigma * 2\,\mathrm{s})\,(\pi 2\mathrm{p}_x)\,(\pi 2\mathrm{p}_y)\,(\sigma 2\mathrm{p}_z)\,(\pi^* 2\mathrm{p}_x)\,(\pi^* 2\mathrm{p}_y)\,(\sigma^* 2\mathrm{p}_z)$$

The valence electrons in C_2, NO, O_2 and N_2 are $8, 11, 12$ and 10 , respectively. The given criteria is satisfied only by NO. Its configuration is

$$(\sigma 2\,\mathrm{s})^2(\sigma * 2\,\mathrm{s})^2\,(\pi 2\mathrm{p}_x)^2\,(\pi 2\mathrm{p}_y)^2\,(\sigma 2\mathrm{p}_z)^2\,(\pi^* 2\mathrm{p}_x)^1$$

In converting NO to NO^{+}, the electron is removed from antibonding molecular orbital and the resultant ion possesses no unpaired of electrons.

Q.12 Answer:

$NaCl$

Solution:

Melting point of $LiCl$ is less than that of $NaCl$ and later it generally decreases on descending the group. The low melting point of $LiCl$ is due to its covalent character.

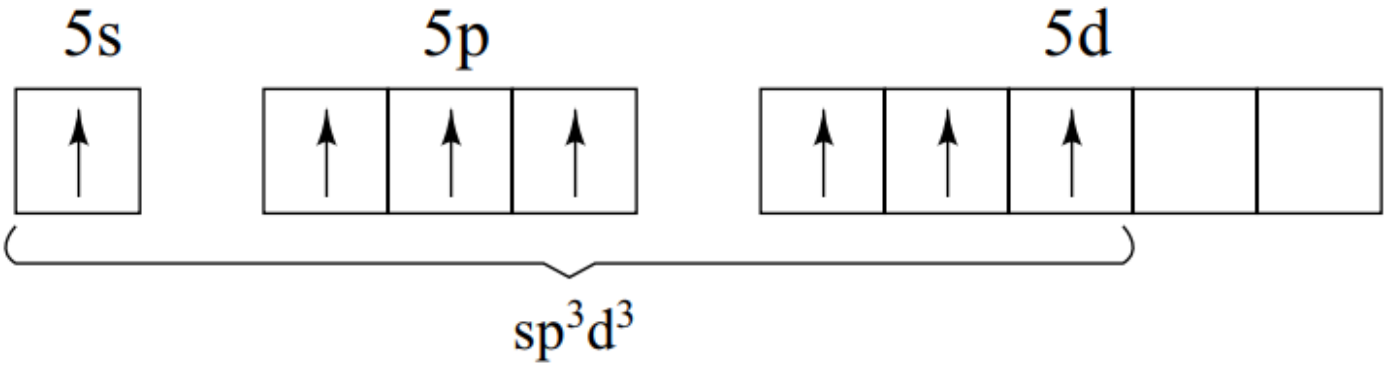
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Q.13 Answer:

$sp^3\ d^3$

Solution:

$sp^3\ d^3$ **hybridization is involved.**



Q.14 Answer:

charge and size of the ions

Solution:

Lattice energy depends on the charge and size of ions.

Q.15 Answer:

$K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$

Solution:

Larger charge and lesser size of the cation leads to larger ratio of charge/size and hence larger polarizing power of the cation.

Q.16 Answer:

$NF_3 < NH_3$ and $OF_2 < OH_2$

Solution:

Bond angle in NF_3 is smaller than in NH_3. Also bond angle in OF_2 is smaller than in OH_2. This is primarily due to ionicity of $N-F$ and $O-F$ bonds. The more electronegative fluorine atoms attract bonding electorn pairs away from N and O nuclei. This causes expansion of lone pair causing bond angles $F-N-F$ and $F-O-F$ smaller than those in $H-N-H$ and $H-O-H$. Experimental values are:
$NF_3\ (102.3^\circ)\ , NH_3\ (107.2^\circ)\ , OF_2\ (103.1^\circ)$ **and** $OH_2\ (104.5^\circ)$

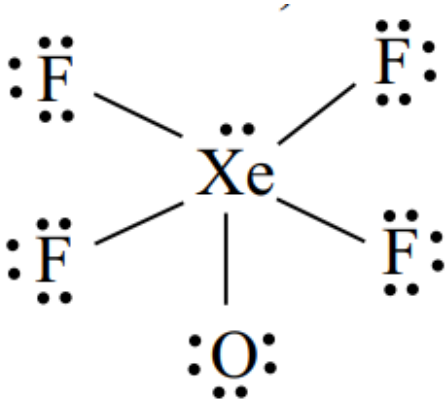
Q.17 Answer:

square pyramidal

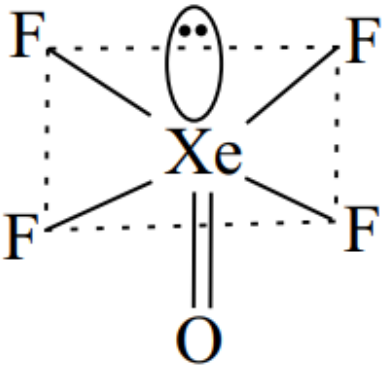
Solution:

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The valence electrons in $XeOF_4$ are $42(=8+6+4\times7)$. These are distributed as follows.



There are 6 pairs of electrons around Xe. Their orientations will be square bipyramidal. The geometry of $XeOF_4$ is square pyramid as the lone pair of electrons occupy axial position



Q.18 Answer:

Hybridization of orbitals predicts the molecular geometry of the molecule and not vice-versa.

Solution:

Molecular geometry must be known to predict the type of hybridization in the molecule.

Q.19 Answer:

$O_2^+ > O_2 > O_2^- > O_2^{2-}$

Solution:

Bond Dissociation Energy $\uparrow \rightarrow$ **Bond Order** $\uparrow$

Bond Order $(B.O) = \frac{1}{2}x\{[\text{No. of } e^{-1} \text{ in Antibonding M.O.}]\text{- } (\text{No of } e^- \text{ in bonding M.O. }]\}$

For O_2 **molecule, electronic configuration is**

$O_2 = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \Pi 2p_x^2 = \Pi 2p_y^2, \Pi^{*2} p_x^1 = \Pi^* 2p^1 y$

B.O. $\cdot (O_2) = \frac{10-6}{2} = 2$

(a) For O_2^+ **molecule, an electron is removed from** $\pi^* 2py$ **orbital**

B.O. $\left(O_2^+\right) = \frac{10-5}{2} = 2.5$

(b) B.O. $(O_2) = 2$

(c) B.O. $\left(O_2^-\right)$ **:electron is added to** $\pi^* 2px^1 y$

$$B \cdot O \cdot \left(O_2^2\right) = \frac{10-7}{2} = 1.5$$

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(d) For O_2^{2-} molecule, two electrons are added to π^*2p^1x ant π^*2p^1y

B.O. $\left(O_2^{2-}\right) = \frac{10-8}{2} = 1$

∴ The order is:

$$O_2^+ > O_2 > O_2^- > O_2^{2-}$$

Higher B.O, Higher Dissociation Energy

Q.20 Answer:

The order of energies of molecular orbitals in F_2 molecule is

$$E(\sigma 2\,s) < E\,(\sigma^* 2\,s) < E\,(\pi 2p_x) = E\,(\pi 2p_y) < E(\sigma 2p) < E\,(\pi^* 2p_x) = E\,(\pi^* 2p_y) < E\,(\sigma^* 2p)$$

Solution:

1. Be_2 has bond order zero, so it is not stable molecule,

2. He_2 has bond order zero and He_2^+ has bond order 1

3.Bond strength of N_2 is maximum as the bond order is 3 that is maximum amongest the homonuclear diatomic molecules belonging to the the second period.

4.The order of energies of molecular orbitals in F_2 molecule is $\sigma 2\,s < \sigma * 2\,s < \sigma 2p_z < (\pi 2p_x = \pi 2py_y) < (\pi * 2p_x = \pi * 2p_y) < \sigma * 2p_z$

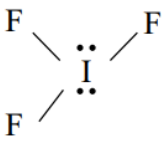
Q.21 Answer:

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Solution:

IF_3

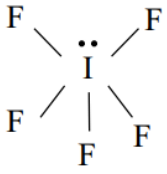
Valence electrons = $4 \times 7 = 28$



(2 pairs)

IF_5

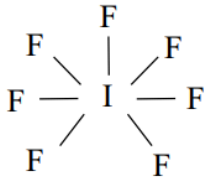
Valence electrons = $6 \times 7 = 42$



(1 pair)

IF_7

Valence electrons = $8 \times 7 = 56$



(0 pair)

=2+1-0=3

Q.22 Answer:

—

Solution:

Valence electron of $O_2^{2-} = 18$

Molecular Orbital Electronic Configuration

$\sigma_{1s}^2\sigma_{1s^*}^2\sigma_{2s}^2\sigma_{2s*}^2\sigma_{2p}^2\pi_{2p}^4\pi_{2p*}^4\sigma_{2p*}^0$

∴ Bonding electron $= 10$

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and Anti-bonding electron $= 8$

$\therefore BO - ABO = 2$

Q.23 Answer:

—

Solution:

Valence electrons in CN^- $= 10$ **Electronic configuration** $(\sigma 2\,\mathrm{s})^2(\sigma * 2\,\mathrm{s})^2\,(\pi 2\mathrm{p}_x)^2\,(\pi 2\mathrm{p}_y)^2\,(\sigma 2\mathrm{p}_z)^2$

Bond order $= \frac{(\text{ Bonding} - \text{antibonding) electrons}}{2} = \frac{8-2}{2} = 3$

Q.24 Answer:

—

Solution:

Per cent ionic character

$$= \frac{\left(1.03 \times 3.3356 \times 10^{-30} \mathrm{Cm}\right)}{\left(1.602 \times 10^{-19}\ \mathrm{kg}\right)\left(127 \times 10^{-12}\ \mathrm{m}\right)} \times 100 = 16.9\%$$

Q.25 Answer:

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Solution:

Since $p = qr$**, we have**

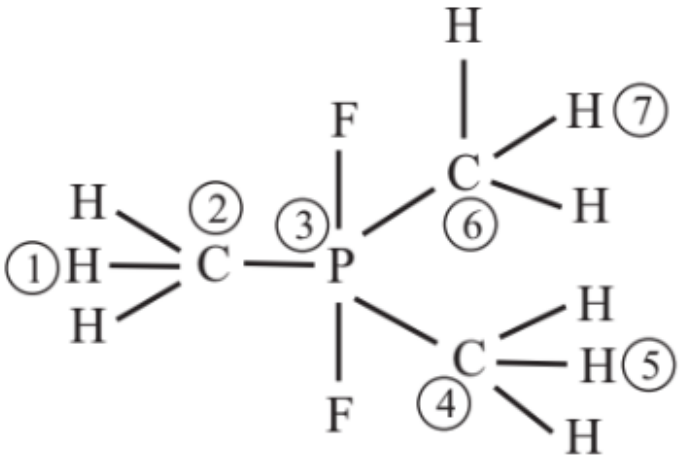
$$q = \frac{p}{r} = \frac{0.38 \times 10^{-18}\ \text{esu cm}}{1.617 \times 10^{-18}\ \text{cm}} = 2.35 \times 10^{-11}\text{esu}$$

Fractional charge on each atom $= \frac{2.35 \times 10^{-11}\ \text{esu}}{4.802 \times 10^{-10}\ \text{esu}} = 0.049$

Q.26 Answer:

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Solution:



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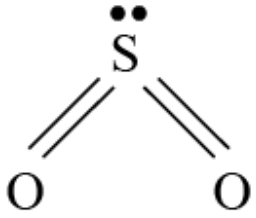
Q.27 Answer:

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Solution:

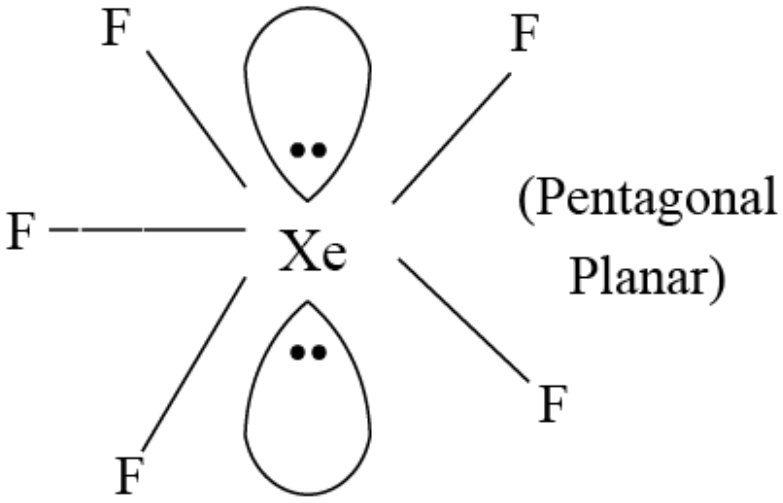
(I) I_3^- **has linear structure. So, the given statement is false.**

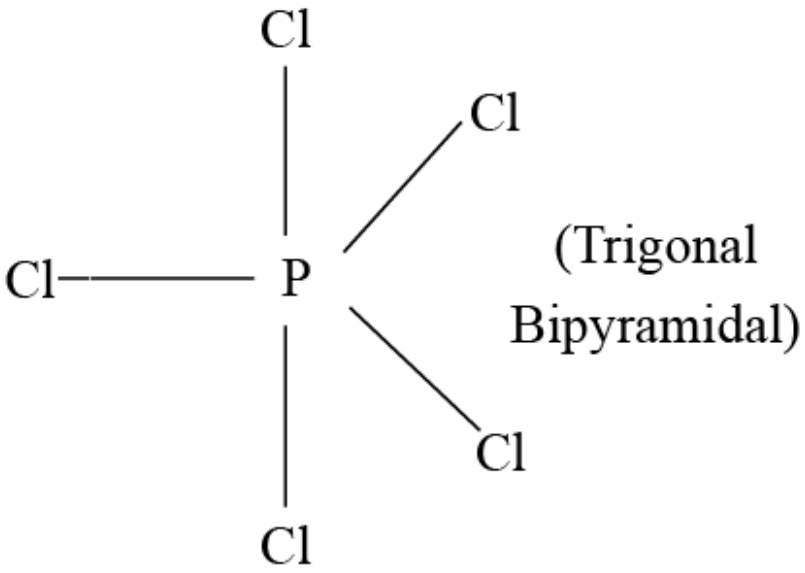
(II)



sp^2 **- Out of 2 unpaired electrons, one forms** $p\pi - p\pi$ **double bond and the other forms** $d\pi - p\pi$ **bond. So, the given statement is correct.**

(III)





So, the given statement is correct.

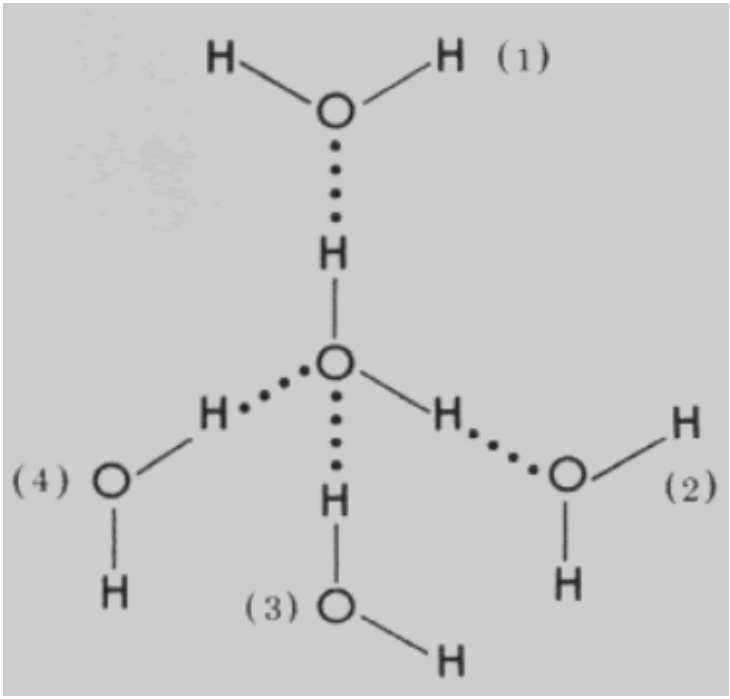
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Hence, 2 statements are correct.

Q.28 Answer:

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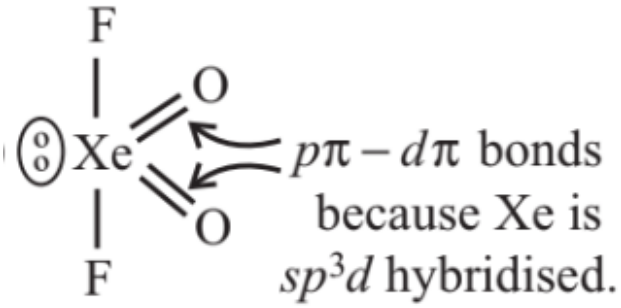
Solution:



Q.29 Answer:

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Solution:



Q.30 Answer:

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Solution:

Per cent ionic character $= \frac{5.8\times10^{-30}\text{Cm}}{(1.602\times10^{-19}\text{C})(145\times10^{-12}\text{ m})} \times 100 = 25\%$

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